

Soumyodip Bhattacharya

sbsb96745@gmail.com | [LinkedIn](#) | [GitHub](#)

Bethuadahari Station Road, Nadia, West Bengal

SKILLS

AI & LLM Engineering: LangChain, LangGraph, RAG (Retrieval-Augmented Generation)

Machine Learning: Scikit-Learn, Pandas, NumPy, Matplotlib, Seaborn

Web & Databases: Next.js, Flask, Express.js, Tailwind CSS, Node.js, SQL, PostgreSQL, Drizzle ORM, SQLAlchemy

Languages: Python, C++

WORK EXPERIENCE

AI Engineer Intern | PreciQube

Jan 2026 – May 2026

- Designed and deployed agentic AI pipelines using LangChain with retrieval-augmented generation (RAG), enabling LLMs to reason over domain-specific knowledge bases — directly relevant to enterprise AI agent development.
- Integrated Anthropic Claude and Google Gemini APIs; engineered structured prompt strategies that improved output reliability and reduced hallucinations for specialised technical domains.
- Delivered end-to-end production Python modules covering API integration, knowledge retrieval, and response validation — gaining practical experience across the full AI deployment lifecycle.

PROJECTS

Medi-Chat: Telehealth & Consultation Platform | Next.js, TypeScript, PostgreSQL, Drizzle ORM

Link: <https://github.com/Soumyo321/ai-medical-app1>

- Engineered a multi-step agentic diagnostic flow (15-question reasoning chain) with dynamic specialist routing — a practical example of structured LLM reasoning over user context.
- Built scalable database architecture with Drizzle ORM for persistent user session and conversation history management.

AI Medical Diagnosis & Recommendation System | Python, Scikit-Learn, Pandas, Flask

Link: <https://github.com/Soumyo321/Medical-Recommendation-System>

- Built an end-to-end intelligent agent that maps patient symptoms to diagnoses and recommendations; achieved 96.43% accuracy via supervised classification on curated medical datasets.
- Constrained model outputs to a fixed taxonomy sourced from verified medical datasets, eliminating open-ended hallucinated responses; structured output into diagnosis, precautions, and medication categories for transparency.

AI-Powered Smart Building Management System | Honeywell Campus Connect Hackathon | Python, EnergyPlus, Groq/Llama 3.3-70B, Streamlit, SQLite

Link: https://github.com/Soumyo321/honeywell_BMS

- Built a full-stack simulation system in a 48-hour solo hackathon, integrating real EnergyPlus 23.2.0 energy modeling with an LLM agent for building optimization recommendations.
- Designed a Streamlit dashboard with SQLite logging for real-time monitoring and decision tracking; delivered complete GitHub documentation and slide deck for submission.

HACKATHONS

- Blostem Hackathon 2026 — Multilingual FD Assistant: Built a multilingual chatbot (Hindi, English, Bengali, Telugu) using LangGraph + Groq (Llama 3.3-70B) to simplify fixed-deposit jargon and guide an 8-step FD booking flow; deployed live on Vercel + Render.
- Flipkart Gridlock 2.0 Hackathon — AI Parking Enforcement System: Built an AI-driven parking intelligence system with a 3-member team using K-Means clustering to detect violation hotspots and generate patrol schedules, with a Flask API and React dashboard.

EDUCATION

B.Tech Computer Science (CSE Core) — Vellore Institute of Technology

CGPA: 8.71

BS Data Science — Diploma Level — IIT Madras

CGPA: 7.09

Class XII — Bethuadahari JCM High School

93.2%

Class X — Bethuadahari JCM High School

92.28%

CERTIFICATIONS

- Data Science, Foundation Level — IIT Madras BS Program
- Coursera -ML and Python Certification
- PreciQube Internship Certificate (AI Engineer)